

CODE SPORT



In this month's column, we continue to discuss the problem of classifying insincere questions on a community question-answering site.

In my last column, we had started discussing the task of classifying questions on community question answering (CQA) forums that appear to be genuine requests for information, but instead aim to convey an opinion and often include hate speech or inappropriate language. We posed this problem in the context of the well known CQA forum, Quora, which has released a data set on Kaggle titled *Insincere Questions*, available at <https://www.kaggle.com/c/quora-insincere-questions-classification/data>.

Given a question, the task is to label it as insincere or not. This is a binary classification problem. Given that CQA questions are typically short, this can also be considered a short text classification problem. In last month's column, we had proposed a simple baseline classifier using LSTM for this task. However, we had pointed out that 'insincere questions' exhibit widely different characteristics. Though this task is modelled as a simple binary text classification problem, the insincere questions arise due to multiple reasons and hence fall under different categories. These include rhetorical questions (that express an opinion), disparaging or offensive questions (that are inflammatory in nature, or are intended to offend a person or group), hypothetical questions and those with objectionable or sexual content.

Since each category of insincere questions is extremely different from others, it may be better to explore whether we can design

separate classifiers for each of these categories and then combine their decisions. Hence, let's first look at the problem of designing classifiers for categorising each of these sub-categories. Rhetorical questions try to express an opinion as a question. Here, the person posting the question is not seeking information, but merely rephrases his or her opinion as a question. For example, consider the following two questions.

Q1: Has the United States become the largest dictatorship in the world?

Q2: How much more political fumbling will it take for Republicans to turn on Trump?

Neither Q1 nor Q2 contain requests for information. They are expressing opinions phrased in a question form. On the other hand, consider the following questions.

Q3: Is the United States the largest grower of coffee in the world?

Q4: How long will it take for the Discovery spacecraft to reach beyond the solar system?

The nature of Q3 and Q4 is similar to Q1 and Q2. However, Q3 and Q4 exhibit actual information-seeking behaviour. Those who have posed Q3 and Q4 have the intent of seeking information whereas the intent of those posting Q1 and Q2 is to express an opinion. Hence, we need a way of distinguishing between these two types of questions — information-seeking questions and non-information-seeking questions. The latter are typically about expressing an opinion, whereas information-seeking questions are posed by those looking for answers or facts. Therefore, non-